

31338 Network Servers

Lab 1a: Virtual Machine Setup using VMWare Workstation Pro

Aims

1. Explore the VMware software environment.
2. Understand how to start and shut down a VMware image.

Task 1: VMware Preparation

Preparation

1. Confirm that the host computer has sufficient memory. At least 8 GB of RAM is required, and 16 GB or more is recommended. This check is not required when using a lab desktop.
2. Register for a Broadcom account and sign in to the Broadcom dashboard. Under *My Downloads*, select *VMware Cloud Foundation*.
3. Search for *VMware Workstation*, then select *VMware Workstation Pro*.
4. Select *VMware Workstation Pro for Personal Use (Windows)* if you are using a Windows laptop or desktop computer.
5. Accept the terms and conditions so that the download button becomes available.

Importing the virtual machine

1. Open VMware and import the .ova file.
2. Choose *Open a Virtual Machine*, select your .ova file, and review the import dialog.
3. Provide a clear virtual machine name and choose a location, then start the import.

Access details

After the virtual machines have been imported, they can be started and used for the lab activities.

- username: SuperUser (CentOS) or Administrator (Windows)
- password: student123!

Task 2: Navigate Through the VMware Lab

You probably need a portable drive for the virtual OS. A USB drive or SSD with more than 80 GB of available space is recommended. For the cross-platform usage, the USB drive or SSD should be formatted to exFAT.

On the lab desktop with Redhat Linux, open *Applications -> System Tools -> VMware Workstation* (or *VMware Player*).

Select *Open a Virtual Machine*, choose the CentOS or WindowsServer .ova file, provide a virtual machine name and storage path, and import the image.

Example storage paths include:

```
/class/NetServers/home/<your student number>/centos64
```

This option applies to the LAN lab only.

```
/media/<USB drive or SSD Label>/centos64
```

The import process normally needs to be completed only once.

Starting VMware systems

After the systems is imported, select *Power on this virtual machine* to start the virtual OS.

If you move to another computer with your USB drive or SSD, open VMware Player, choose *File -> Open*, browse to the relevant directory on the external drive, and select the .vmx file for the virtual machine. Usually there will be only one .vmx file.

The startup panel should display details such as the system state, operating system, version, and allocated RAM.

Select *Edit VM settings* and review the listed devices. Consider what each device represents and which settings a system administrator may reasonably change.

System administrators are expected to document system configurations. Record the configuration details in your learning journal.

In most organisations, hardware and configuration records are maintained through formal processes such as enterprise configuration management tools, databases, spreadsheets, or paper-based records. In this lab, your learning journal provides that record.

To give the VM mouse and keyboard focus, click inside the virtual machine window.

Hint: when using the console, press *Ctrl+Alt* to return control to the desktop.

Observe the following during startup:

1. The BIOS startup screen appears briefly. Optional activity: to view the setup screen, press F2 as soon as the logo appears. Review settings such as the date, time, and boot sequence.

2. When the GRUB logo appears, either allow the default Linux system to boot or press a key quickly to open the GRUB menu. The GRUB menu allows startup parameters to be changed. Consider which parameters are visible in the Linux configuration.
3. When the system begins booting, a progress bar appears at the bottom of the screen. Press Esc to display the boot sequence. For this activity, focus on observing the Linux startup process.

Task 2: Logging in as SuperUser (root) and Shutting Down

Log in as the root user in CentOS 10, also known as the superuser. When prompted for a password, enter

```
student123!
```

This root password is deliberately simple for the disposable lab environment. It would be unacceptable for a real system.

The gnome-initial-setup process may ask about language, keyboard settings, location services, and online accounts. Do not connect personal online or social media accounts.

You may explore the system briefly, but do not change the password.

Open a terminal, and check the network configuration by entering:

```
ifconfig
```

What information is displayed? Can you determine whether a default TCP/IP address has been assigned?

Recall the Unix commands introduced in Web Systems or Unix Systems Programming. For example, try the `ls` command and observe the output.

Logging on as root

Logging in directly as root is rarely appropriate in normal administration. The trivial password is used only because this is a disposable lab system. In a real environment, the root password should be difficult to guess or crack because root has unrestricted administrative power. Work carefully, because it is easy to damage an installation when operating as root.

The best practice is to log in with a personal account and use `sudo` for privileged commands. This will be examined later. For now, the next best approach is to use the superuser command:

```
su
```

Enter the password when prompted. Notice that the prompt changes to the `#` symbol, which indicates superuser privileges.

Rebooting and shutting down

There are several commands for rebooting a system. In this context, the simplest is:

reboot

Why would a system administrator not normally reboot a server without careful planning? Observe the screen during the shutdown and reboot process, and record relevant notes in your learning journal.

When ready, shut down the virtual machine with:

```
shutdown -h now
```

Again, document what happens.

The Linux distribution used in this lab also supports shutdown or restart through the graphical menus. Select the power icon near the top right of the screen to explore this option.

Why is it important to know how to shut down a system from the command line? Record your answer in your learning journal.